

Skill assessment — what to build, in what order

Hand this to Replit alongside the ten screenshots listed at the end. Every screen referenced here exists as a clickable screen in [Skill Assessment — Foundations Trail](#); that document is the visual spec and this one is the instruction. Where the two disagree, the screens win.

Nonprofit Cloud · Agentforce · Gemini 2.5 Flash

10 screens in scope

4 phases

US spelling throughout

1 · The instrument in one paragraph

A learner sits an adaptive-length assessment three times — at onboarding, at Week 6, and at program end — drawn from one item bank at the published Salesforce Administrator domain weights. Each domain stops when its read stabilizes rather than at a fixed count, so a sitting is roughly 32 to 58 items. Three item types: multiple choice, a written scenario scored against three fixed criteria, and a build verified by reading the learner's own dev org. Every item also captures how sure the learner was. Alongside the three official sittings, practice is unlimited, coached by Penny with immediate feedback, unscored, and any item seen in practice is retired from that learner's next official draw. The coach records an independent judgment of the learner before the read is visible to them, which is our only evidence that the instrument measures something a person would recognize.

2 · Decisions already made

These are settled and each one has a reason. If a build constraint makes one impossible, raise it rather than quietly choosing the other way — every one of them is load-bearing for a claim we make to funders.

No total at intake	Per-domain reads only	A weighted total at intake tells someone they are failing a thing they have not started, and a first score can end a program. The weighted estimate against the 65% pass mark appears at Week 6, once there is movement beside it.
Confidence visibility	Coach only, never the learner	Confident-and-wrong is a misconception; guessing-and-right is luck. Naming either back to the learner reads as a correction. The coach view separates them.
Penny during official sittings	Silent proctor, no coaching topic	Not a disabled button — no coaching topic is enabled at all during the three sittings, so there is nothing to jailbreak. She is a timer and a save indicator.
Practice results	Is_Practice__c flag, walled off from reporting	One write path and one place items are answered, with the flag set inside TT_WriteAssessmentResult and nowhere else. Then no report type over practice rows and a separate permission set — a filter cannot be forgotten if no report exists to forget it in.
Item exposure	Seen-by-learner junction, practice items retired	Any item shown in practice is withdrawn from that learner's next official draw. Without this, practice quietly teaches the measure and measured gain becomes recall.
Domain weighting and thresholds	Item bank record, never code	The exam re-weighted in December 2025 and will again. A weighting held in a record survives that and stays traceable; one held in code does not.

Proctoring scope	Tab-only share, stills every 15s	We ask for this tab, not the whole screen. That costs us the one signal that could name where someone went, and we accept it — the learner's note and dev-org login history carry attribution instead.
Proctoring retention	30 days, job before first upload	Long enough for the review conversation, short enough that a breach or a records request finds almost nothing. The delete job ships before the first frame is stored, not after.
Coach rating order	Rate before the read opens, enforced on the record	Rate afterwards and we have measured whether a coach can read a number. Both timestamps stored; the row is read-only after insert.

3 · Data model

Nonprofit Cloud. Two of these exist today; the rest are new. Create the picklist values before anything writes to them, and treat every threshold and weight as data on the item bank record rather than a constant in code — the exam re-weighted in December 2025 and will again.

Assessment__c (on Program Engagement)

Exists, needs fields

Add: Occasion__c (intake / midpoint / exit), Item_Bank_Version__c, Proctoring__c (captured / declined / unavailable), Drive_Folder_Id__c

Exists in production and is absent from the data model workbook — start by documenting what is already there.

TT_Assessment_Result__c

Create

Sitting, Domain__c, Mode__c (mc / scenario / build), Correct__c, Confidence__c (restricted picklist, 3 values), Is_Practice__c, rubric criterion fields, provenance fields (keystrokes, paste events, longest insertion, pasted share, derived flag)

One row per item answered. Create the picklist values before anything writes. Store criteria met, never a prose summary — a summary cannot be reported on.

TT_Coach_Rating__c [Create](#)

Sitting, Domain__c, Band__c (Emerging / Proficient / Exemplary / Not observed), Rated_At__c, Read_Opened_At__c

One row per domain per sitting. Read-only after insert, and the gate enforced on the record rather than in the UI.

TT_Focus_Event__c [Create](#)

Sitting, Offset_Seconds__c, Duration_Seconds__c, Frame_Captured__c, Learner_Note__c, Note_Declined__c

One row per switch away. Store events, never a total — four ten-second checks read differently from one four-minute absence.

Item bank (item records) [Create](#)

Domain, weight, mode, correct answer, three rubric criteria, accepted synonyms, build checks list, statistics rollups (response count, correct rate, discrimination, confident-and-wrong rate), retirement version

The whole configuration surface. Statistics roll up from stored responses and cannot be backfilled — build the rollups in Phase 1 even though nothing reads them until Phase 3.

Item exposure junction [Create](#)

Learner, item, first seen occasion (official / practice)

What makes practice safe. The draw action excludes anything in here for that learner.

Google shared drive [Create](#)

/Trail OS Assessments/{username}/{assessment type}/{YYYY-MM-DD}/frame-NNNN-SSSS.jpg

Org-owned shared drive, never a personal My Drive. Folder keyed on Salesforce username with the Contact id in appProperties so a rename does not orphan history. Salesforce stores the file id and link only.

Course_Module_Activity__c [In production](#)

Quest_Eligible__c

Penny's recommended next step must resolve to a real curriculum activity, so the step is trackable rather than advice.

LoginHistory (learner dev org)

In production

Read via the learner's org connection

Not ours to create — queried at review time to confirm the one destination we can verify independently.

4 • The Agentforce topic

One topic, eight named actions. Penny never touches an object directly — she calls these, so every read and write has a signature and a log entry. Agentforce orchestrates; Gemini reasons only where language is the point.

TT_DrawAssessmentItems

Apex invocable

In: program engagement, occasion, item bank version · Out: ordered item ids per domain

Draws at the published weights and excludes the exposure junction. Reproducible from the record — no model involvement.

TT_ScoreObjectiveItem

Flow

In: item id, response · Out: correct, domain

Compared to the correct answer on the item record. No model in this path, ever.

TT_EvaluateRubric

Prompt template + Apex

In: item id, written response, three criteria · Out: criterion met per criterion, evaluator confidence

Grounded on the item record only. Low evaluator confidence routes to the coach instead of writing a verdict.

TT_VerifyOrgBuild

Apex invocable

In: dev org connection, checks list from the item record · Out: checks passed, failed, unmatched labels

Describe calls, Tooling SOQL for active flow versions and report folders, Setup Audit Trail for when and by whom. Read scope only. Unmatched labels go to the coach, not to a fail.

TT_LogFocusEvent

Apex invocable

In: sitting, offset, duration, frame captured, learner note · Out: focus event id

Fires on return so a duration exists. Never blocks the sitting, never notifies the learner.

TT_ResolveDriveFolder

Apex · named credential

In: username, assessment type, sitting date · Out: folder id

Creates missing levels by parent plus name, idempotently. Two frames in the same second must not create two folders.

TT_StoreProctorFrame

Apex invocable

In: sitting, JPEG still, elapsed offset, folder id · Out: Drive file id and link

Queues and retries if Drive is unreachable rather than failing the sitting. No bytes in Salesforce storage.

TT_WriteAssessmentResult

Apex invocable

In: sitting, domain, mode, correct, confidence, practice flag, provenance · Out: result id

The only place a result is written and the only place the practice flag is set.

5 · Guardrails that have to live in code

Not policy documents. Each of these is a thing a future change could quietly break, so each needs to be enforced where it cannot be forgotten.

Scoring is never a model judgment

The correct answer and the rubric criteria live on the item record. A model that is 94% reliable is a broken exam. Gemini reads language; it never decides whether an answer is right.

No coaching topic during official sittings

Enforced by topic configuration per occasion, not by hiding a chat box. Nothing to jailbreak means nothing to police.

Everything outbound is human-gated

Coach summaries and any flag naming a misconception are drafts until a human sends them. Nothing about a learner reaches a partner or funder surface except as a name-free aggregate.

Read-only scopes on the learner org

The connected app requests no write scope at all, so a bug in verification cannot damage someone's dev org. Verify this at review, not at incident.

Practice is unreachable from reporting

Flag set in one action, no report type over practice rows, separate permission set. The measure cannot be contaminated by a forgotten filter.

Capture never blocks progress

Declined, unsupported, or unreachable Drive all continue the sitting and write an honest value. A learner must always be able to finish.

6 • Phased plan

Four phases. Each one ends in something a learner or coach can actually use, and each has a test that has to pass before the next begins. Phase 1 is the whole instrument minus the evidence apparatus — if only Phase 1 ever ships, the program still works.

PHASE 1 The instrument

Screens 1–6

- ☐ Item bank objects, the three item types, and the adaptive stop rule reading its threshold from the bank record.
- ☐ Confidence capture on every item, restricted picklist, coach-visible only.
- ☐ TT_DrawAssessmentItems, TT_ScoreObjectiveItem, TT_WriteAssessmentResult, TT_EvaluateRubric, TT_VerifyOrgBuild.
- ☐ Intake read: per-domain only, no total, one next step resolving to a real curriculum activity.
- ☐ Per-item response storage and the statistics rollups — nothing reads them yet, and they cannot be added later.

DONE WHEN

A learner completes a sitting end to end, the domains stop at different item counts, the read shows eight domains and no total, and the item bank shows response counts climbing on the items that were drawn.

PHASE 2 Practice and exposure

Screens 7–8

- ☐ The exposure junction, and draw exclusion for anything the learner has seen.
- ☐ Practice mode: unlimited, learner-initiated, immediate Penny feedback with why each wrong option fails.
- ☐ Is_Practice__c set inside the write action, no report type over practice rows, separate permission set.
- ☐ Practice volume visible to the coach; practice performance visible to nobody.

DONE WHEN

An item answered in practice never appears in that learner's next official draw, and no report in the org can be built that includes a practice row.

PHASE 3 Evidence apparatus

Screens 9–10

- ☐ TT_Coach_Rating__c with the gate on the record: rate all domains, then the read opens. Read-only after insert.
- ☐ Agreement view — exact, adjacent, opposed — per sitting, aggregating to cohort level.
- ☐ Week 6 comparison against intake using the same blueprint, with the weighted estimate appearing for the first time.
- ☐ Item statistics surfaced: difficulty, discrimination, confident-and-wrong rate, and the retirement rule.

DONE WHEN

A coach cannot see a learner's read until their own rating is committed, an attempt to edit a submitted rating fails at the record layer, and the agreement figures compute across a cohort.

PHASE 4 Proctoring and provenance

Screen 3 detail

- ☐ Response provenance: keystrokes, paste events, longest insertion, pasted share, plus the derived flag with its threshold on the bank record.
- ☐ Tab-share capture with the browser prompt as consent, stills every 15 seconds, plus one at submit.
- ☐ Drive folder resolution and frame upload, pointer to Salesforce, queue and retry.
- ☐ Focus events with the one-line learner note on return, and the dev-org login check at coach review.
- ☐ The 30-day retention job — built and scheduled before the first upload.

DONE WHEN

Pasting into a written response moves the coach signal without changing the score or telling the learner; declining the share completes the sitting and writes declined; and a frame older than 30 days is gone without anyone running anything by hand.

7 • What not to build

- ✗ **No total score at intake.** Not collapsed, not behind a toggle, not in an API response the front end could render. It does not exist until Week 6.
- ✗ **No automatic penalty from paste detection or focus events.** These produce signals a human reads. Nothing scores down, nothing locks, nothing accuses.
- ✗ **No blocking on capture.** Declining or an unsupported browser must not prevent a sitting. Write the honest value and continue.
- ✗ **No model in the objective scoring path.** Multiple choice and the stop rule are deterministic. If a prompt appears in that path, something has gone wrong.
- ✗ **No bytes in Salesforce storage.** Frames live in Drive. Salesforce holds the file id and link.
- ✗ **No confidence shown to the learner.** Not in the read, not in practice, not in a summary email.
- ✗ **No unsolicited Penny message.** She answers when asked and speaks at defined moments. She never opens a conversation on her own, and never posts anywhere a client can see.

8 • Context, not work

Three internal screens exist that are not in this build. They are why the fields above are shaped the way they are, and worth ten minutes before starting.

Evidence model

The eight links from one answered item to a funder claim, each marked in production, designed, or missing. Explains why item statistics are Phase 1 work rather than Phase 3.

Cohort view

The name-free aggregate a funder is shown: denominators, intervals, attrition, and rows that say "not yet measured" rather than estimating. Nothing in this build renders it yet.

Item bank health

Difficulty, discrimination, and the retirement rules, with equating readiness per domain. The reason thresholds live in the bank record.

9 • The screenshot set

Ten screens, in this order, captured from the design document. Each line names what to look at rather than what it looks like.

1	Before you start	The four facts framing the sitting — length, three occasions, who sees it, no retakes. Sets the tone the rest inherits.
2	Multiple choice	Item layout, the confidence question as part of the item rather than an afterthought, and the domain rail showing which domains are settled.
3	Scenario item	Rubric shown before writing, the coach signal panel under the box, the session capture panel, and the away-events panel. The densest screen in the build.
4	Build check	Steps, what Penny will check, and the how-each-check-is-verified block naming the actual describe and Tooling SOQL calls.
5	Domains settling	Why the sitting ends when it ends — per-domain stability rather than a fixed count.
6	Your read	Per-domain only. No total anywhere on the screen, and one concrete next step in the heaviest weak domain.
7	Practice home	Domains ordered by where movement is worth the most, unseen-item counts, and the four rules of practice.
8	Practice item	Immediate feedback, why each wrong option fails, and Penny asking one question rather than explaining at length.
9	Week 6 movement	Intake mark against the Week 6 bar, a domain that went down shown honestly, and the weighted estimate appearing for the first time.
10	Coach rating	The gate before submit, then the agreement comparison. The order of these two states is the whole feature.

① **Two things still owed by us, not by Replit —**

The measurement protocol page — success threshold, expected domains, attrition rule, denominator — has to be written before the next intake, or every figure it governs reads as chosen after the fact. And the six- and twelve-month outcome follow-up needs a named owner; nothing in this build asks anyone to collect it, which is exactly how that row stays empty.